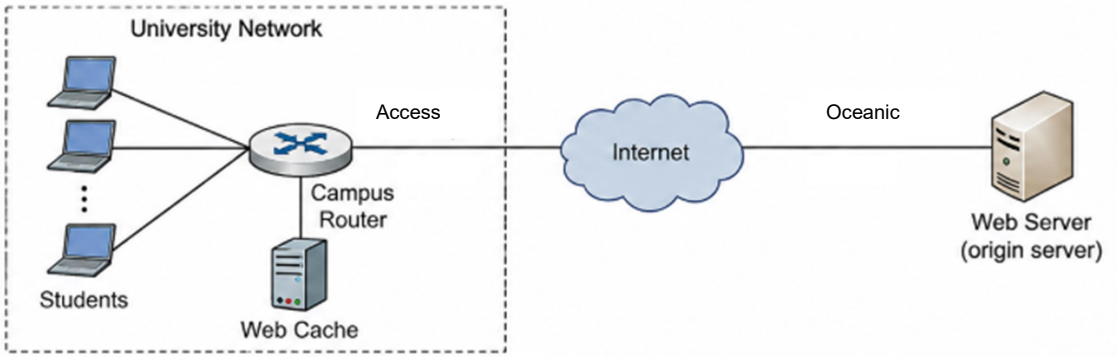
CN mdi1 solution - Fall 2026

Q1: [Estimated Time: 30 min] [0.5 + 1.5 + 1 + 2 + 1 + 1.5 = 7.5 Marks]

Consider the university network shown in the figure. Students access a Web Server through the network, and a local web cache is available at the campus router. The packet size is 12,000 bits. The network has 3 links:

- Oceanic Link: Web Server to ISP router. Length = 10,000 km, Propagation speed = 2×10^8m/s, Link Bandwidth = 100 Mbps.
- Access Link: ISP router to Campus router. Link Bandwidth = 10 Mbps, Propagation delay = 20 ms.
- Campus Link: Campus router to Students. Link Bandwidth = 1 Gbps, Propagation delay = negligible.



Assume no queuing or processing delay.

(a) Identify the bottleneck link in the network and determine the maximum throughput that can be delivered to a single student. [0.5 Marks]

Bottleneck is the minimum capacity link = 10 Mbps [ISP-Campus]. Max rate = 10 Mbps

(b) Calculate the total end-to-end delay experienced by one packet from the Web Server to a student. Separate the transmission and propagation delays. [1.5 Marks]

Propagation:

Oceanic = 10,000km / 2 * 10^8 = 50 ms.

Access = 20 ms.

Campus = ~0 ms.

Total prop = 70 ms._

Transmission:

Oceanic = 12,000 / 100 Mbps = 0.12 ms.

Access = 12,000 / 10 Mbps = 1.2 ms.

Campus = 12,000 / 1 Gbps = 0.012 ms.

Total trans = 1.332 ms._

Total Delay = 70 + 1.332 = 71.332 ms.

(c) The cache currently satisfies 40% of requests. If the request rate is 10 requests/second and each object is 100,000 bits, calculate the data rate utilizing the access link of 10 Mbps. [1 Marks]

Rate = (1 - hit) * req/sec * object size = 0.6 * 10 * 100,000 = 600 Kbps.

(d) A webpage consists of 1 HTML file and 6 images. Assume the HTML file must be retrieved first, after which the images can be retrieved. Assume 1 RTT for TCP handshake and 1 RTT for object transfer. Calculate the total number of TCP connections and RTTs required using: [2 Marks]

(i) Non-persistent HTTP
(ii) Persistent HTTP

(i) Non-Persistent HTTP:

Total 7 TCP connections (one for each of the 7 objects)

Total 14 RTTs (for each of the 7 objects: 1 RTT for TCP connection, and 1 RTT).

(or 7 RTTs if counting transfer only). Accept both if explained.

(ii) Persistent:

1 TCP connection total.

Total 8 RTTs (1 RTT for TCP connection, and for each of the 7 objects: 1 RTT for object transfer)

(or 7 RTTs if counting transfer only). Accept both if explained.

(e) As the webpage data moves through the protocol stack, state the PDU/data-unit name at each layer: Application → Transport → Network → Data Link → Physical. [1 Marks]

Message->Segment->Datagram->Frame->Bits

(f) Regarding Transport Service Requirements, Redraw the table and fill each cell with Yes / No. [1.5 Marks]

Application	**Loss can be tolerated?**	**Needs min. throughput?**	**Delay Sensitive?**
Uploading an image on a webpage	No	No	No
Video chat on Google Meet	Yes	Yes	Yes
Online gaming (voice chat)	Yes	Yes	Yes
Watching buffered video (YouTube)	Yes	Yes	Yes

CLO # 1: Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Q2: [Estimated Time: 25 min] [3.5 + 1 + 1.5 + 1.5 = 7.5 Marks]

(a) A user types www.nu.edu.pk in browser. Browser cache is empty. [3.5 Marks]

(i) List the 4 steps of DNS resolution in order. Use format: Browser -> ... -> ... -> IP returned. [2 Marks]

1.Browser cache / OS cache check

2. Query to Local DNS

3. Local DNS -> Root -> TLD -> Authoritative DNS (Iterative)

4. Authoritative returns IP to Local -> Browser

(ii) Identify the DNS resource record for each function: [0.5*3 = 1.5 Marks]

a. Maps domain name to IPv4 address: _____
b. Creates an alias for another domain: _____
c. Specifies mail server for a domain: _____

a. A record

b. CNAME

c. MX record

(b) Define Spatial and Temporal coding with the help of equation in max. TWO lines. [1 Mark]

Spatial = Removes redundancy within a frame / Intra-frame, e.g., DCT, JPEG-like.

Temporal = Removes redundancy between frames / Inter-frame, e.g., Motion compensation, difference coding

(c) List any two improvements of HTTP/2 over HTTP/1.1 and any one improvement of HTTP/3 over HTTP/2. [1.5 Mark]

HTTP/2:

- transmission order of requested objects based on client-specified object priority (not necessarily FCFS)
- push unrequested objects to client
- divide objects into frames, schedule frames to mitigate HOL blocking

HTTP/3:

- Uses QUIC over UDP, 0-RTT, improved loss recovery

(d) What is stored in the DASH manifest file (.mpd)? How does it help adaptive streaming? Answer in max 2 lines. [1.5 Mark]

Manifest stores list of available bitrates / resolutions and segment URLs / durations. Client measures bandwidth and picks appropriate bitrate for next segment from manifest

***** The End *****